

Answers to three problems of Hak, Kozerenko and Oliynyk on the triameter of graphs

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Abstract

For a connected graph G , the triameter $\text{tr}(G)$ is the largest value of $d(u, v) + d(u, w) + d(v, w)$ over vertices u, v, w . Hak, Kozerenko and Oliynyk asked whether every triametral triple of a median graph contains a peripheral vertex (Problem 1), whether every diametral pair of a median graph extends to a triametral triple (Problem 2), and whether every distance-hereditary graph has at least one of these two properties (Problem 3). For Problem 1 we give an 8-vertex median counterexample, which exhaustive enumeration shows to be the unique smallest one. Problem 2 was answered negatively on MathOverflow in December 2025 with an 11-vertex example; we show that the smallest counterexample has 8 vertices, is unique, and is also distance-hereditary. Problem 3 has a positive answer in a stronger form: in a connected distance-hereditary graph, every triametral triple contains a diametral pair or every diametral pair extends to a triametral triple. The proof uses the four-point condition of Bandelt and Mulder. The counterexamples, and the answer to Problem 3 for graphs that satisfy the four-point condition, are formally verified in Lean 4. The results were found by AI coding agents working under the author's direction; their role is described at the end of the paper.

1 Introduction

All graphs in this paper are finite and simple. For a connected graph G with path metric $d = d_G$, let $\text{diam}(G)$ be its diameter, put $d(u, v, w) = d(u, v) + d(u, w) + d(v, w)$, and let the *triameter* $\text{tr}(G)$ be the largest value of $d(u, v, w)$ over all $u, v, w \in V(G)$ [2]. A pair $\{u, v\}$ is *diametral* if $d(u, v) = \text{diam}(G)$, a vertex is *peripheral* if it belongs to a diametral pair, and a triple u, v, w is *triametral* if $d(u, v, w) = \text{tr}(G)$. Das [2] asked whether, in a tree, every triametral triple contains a diametral pair and every diametral pair extends to a triametral triple. Hak, Kozerenko and Oliynyk [6] proved both statements for all connected block graphs [6, Theorem 1.2], showed by examples that they fail for other classes, introduced weaker versions, and posed three problems [6, Section 4]. In our notation, a connected graph G has property (Q3) if every triametral triple of G contains a diametral pair; (Q3') if every triametral triple of G contains a peripheral vertex; (Q4) if every diametral pair $\{x, y\}$ of G extends to a triametral triple, that is, $d(x, y, z) = \text{tr}(G)$ for some vertex z ; (Q4') if every peripheral vertex of G belongs to a triametral triple. These are Questions 3, 3', 4 and 4' of [6]. We restate the three problems of [6, Section 4] in this notation.

Problem 1. Does every median graph have property (Q3')?

Problem 2. Does every median graph have property (Q4)?



Figure 1: The graphs G_1 and G_2 at the grid points of Section 3. Squares are peripheral vertices. In G_1 the white vertices 1, 5, 6 form a triametal triple without a peripheral vertex; in G_2 the white vertices 1, 4, 7 form the only triametal triple.

Problem 3. Does every distance-hereditary graph have property (Q3') or property (Q4)? Both (Q3') and (Q4) hold for trees [6, Theorem 1.2] and for hypercubes, in which every vertex is peripheral and every pair of vertices extends to a triametal triple [6, Corollary 2.5]; and each of them fails for some distance-hereditary graph [6, Figure 3].

Results. Our examples for Problems 1 and 2 are the median graphs G_1 and G_2 on the vertex set $\{0, \dots, 7\}$ in Figure 1; the labels are those of the Lean formalisation.

Theorem A. *Let G_1 be the graph with the ten edges 01, 02, 04, 06, 13, 23, 25, 45, 47, 67. Then G_1 is a median graph with $\text{diam}(G_1) = 4$ and $\text{tr}(G_1) = 8$, its only peripheral vertices are 3 and 7, and $\{1, 5, 6\}$ is a triametal triple. So G_1 does not have property (Q3'), and the answer to Problem 1 is negative. Moreover, every median graph with at most seven vertices has property (Q3'), and G_1 is, up to isomorphism, the only median graph with eight vertices that does not.*

Theorem B (a negative answer to Problem 2 was first given in [13]). *Let G_2 be the graph with the eight edges 02, 03, 06, 12, 25, 34, 35, 67. Then G_2 is a median graph and a distance-hereditary graph with $\text{diam}(G_2) = 4$ and $\text{tr}(G_2) = 12$, and $\{1, 4, 7\}$ is its only triametal triple. So the diametral pair $\{5, 7\}$ does not extend to a triametal triple, and the peripheral vertex 5 belongs to no triametal triple: G_2 has neither property (Q4) nor property (Q4'). Moreover, every median graph with at most seven vertices has property (Q4), and G_2 is, up to isomorphism, the only median graph with eight vertices that does not have (Q4), and also the only one that does not have (Q4').*

Theorem C. *Let G be a connected distance-hereditary graph, and let $\{a, b, c\}$ be a triametal triple of G that contains no diametral pair. Then for every diametral pair $\{x, y\}$ of G ,*

$$\max\{d(a, x, y), d(b, x, y), d(c, x, y)\} = \text{tr}(G).$$

Consequently every connected distance-hereditary graph has property (Q3) or property (Q4), and hence property (Q3') or property (Q4). The answer to Problem 3 is positive.

The last sentences of Theorems A and B rest on exhaustive enumerations (Section 4). Theorem C holds for all finite connected graphs that satisfy the four-point condition of Bandelt and Mulder [1, Theorem 2] (Section 5); Remark 4 shows that it is sharp. The counterexamples G_1 and G_2 , and Theorem C under the four-point condition, are formally verified in Lean 4 (Section 7).

Prior work. The negative answer to Problem 2 is not new. Hak and Kozerenko restated the question in the Lviv Scottish Book [5] (volume 3, page 154, entry dated 10 February 2025), asking whether (Q4), or at least (Q4'), holds for median graphs; the entry also suggests asking whether (Q3') does, which is Problem 1. The problem was relayed to MathOverflow as question 506431 [9] on 28 December 2025, and on 31 December 2025 the user rgvalenciaalbornoz answered it [13], in an accepted answer, with an 11-vertex median graph made of four 4-cycles in which both (Q4) and (Q4') fail. That answer does not concern (Q3'). Our contribution to Problem 2 is only the smallest counterexample of Theorem B, and the observation that it is also distance-hereditary.

For Problems 1 and 3 we found no earlier answer. We searched on 26 September 2026 between 06:43 and 07:21 UTC: the arXiv (its API, for triameter, triametral, triametric and for the papers of Kozerenko); the works citing [6] listed by OpenAlex and Semantic Scholar, read in full where accessible; Crossref; zbMATH Open; MathDB; MathOverflow and Mathematics Stack Exchange, through the Stack Exchange API; the Lviv Scottish Book; Wikipedia; MathWorld; GitHub (code, commits, issues, pull requests, repositories); the repositories `trureturing` [15], `formal-conjectures` [4] and `SCOPE2026` [14] of AI-assisted work on open problems; A. Hak's bachelor thesis (2022); and web search. The Wikipedia article on the triameter [16], last revised on 19 October 2025, before the MathOverflow answer, lists Problem 2 as open and suggests studying (Q3') and (Q4') for median graphs; the paper [8] (29 August 2026) mentions such questions for median and distance-hereditary graphs without answering them.¹ Some searches failed: the keyword searches of OpenAlex and Semantic Scholar returned HTTP error 429 (rate limit), and we used keyword searches of Crossref instead; one citing paper, in *Symmetry*, was judged from its abstract, as its full text was not accessible to us; and a citation search in zbMATH Open gave no usable result. We did not search Google Scholar, Scopus, Web of Science or ResearchGate. General web search did not find the MathOverflow thread, which we found only through the Stack Exchange API. A final check on 26 September 2026 between 15:49 and 15:52 UTC (the arXiv, GitHub, MathDB and MathOverflow) found nothing new. This reports the coverage of our searches, not a guarantee of priority.

We quote [6] from its arXiv version 1; its theorem, figure and problem numbers refer to that version.

2 Definitions and readings

Let G be a connected graph with path metric d and $D = \text{diam}(G)$. A vertex u is peripheral if and only if its eccentricity $\max_v d(u, v)$ equals D . As in [6], the vertices in the definition of $\text{tr}(G)$ need not be distinct. For a diametral pair $\{x, y\}$ and any vertex z , the triangle inequality gives $d(x, y, z) \geq 2d(x, y) = 2D$; hence $2D \leq \text{tr}(G) \leq 3D$ [6, Proposition 2.1]. For vertices u, v let $[u, v] = \{x : d(u, x) + d(x, v) = d(u, v)\}$. A connected graph is a *median graph* if $|[u, v] \cap [u, w] \cap [v, w]| = 1$ for all vertices u, v, w [11]; the unique vertex is the *median* of u, v, w . A connected graph is *distance-hereditary* if every connected induced subgraph H is isometric, that is, $d_H(u, v) = d_G(u, v)$ for all $u, v \in V(H)$ [7].

Readings. A triple with a repeated vertex has $d(u, u, v) = 2d(u, v) \leq 2D \leq \text{tr}(G)$, with equality only if $\text{tr}(G) = 2D$ and $\{u, v\}$ is diametral; so allowing repeated vertices changes nothing in (Q3) and (Q3'). In (Q4), the vertex z may be required to differ from x and y (the *strong reading*) or not (the *weak reading*). If G has at least three vertices the readings agree: if some $z \in \{x, y\}$ works, then $\text{tr}(G) = 2D$, and then every z works; similarly for (Q4'). Our counterexamples refute every reading, and Theorem C holds under the strong one. Also, (Q4)

¹A figure caption in [16] calls the graph G of [6, Figure 3] ($K_{2,3}$ with a pendant vertex) a median graph. It lacks (Q3'), but it is not median, as its vertices a, b, c have two medians; so it does not answer Problem 1.

implies (Q4'), since a peripheral vertex x lies in a diametral pair, which extends to a triametral triple containing x .

The sentence of [6] that introduces Problem 2 describes (Q3), while the problem names Question 4. As [6, Figure 4] already shows that (Q3) fails for median graphs, and the restatement [5, 9] concerns (Q4), we read Problem 2 as stated (G_1 also lacks (Q3)). Problem 3 could be read for the class (all distance-hereditary graphs have (Q3'), or all have (Q4)), but the two graphs of [6, Figure 3], shown just before the problem, refute this reading; so we read it for each graph. As both vertices of a diametral pair are peripheral, (Q3) implies (Q3'); so Theorem C gives more than a positive answer, and it also gives (Q3') or (Q4').

3 Two median graphs

We prove Theorems A and B, except for their last sentences, by placing G_1 and G_2 in the integer grid. For $u, v \in \mathbb{Z}^k$ let $\|u - v\| = \sum_i |u_i - v_i|$, and for integers a, b, c let $\text{med}(a, b, c)$ be the middle one of the three. For $S \subseteq \mathbb{Z}^k$ let $\Gamma(S)$ be the graph on S in which u and v are adjacent when $\|u - v\| = 1$.

Lemma 1. *Let $S \subseteq \mathbb{Z}^k$ be finite, and suppose that (a) $d_{\Gamma(S)}(u, v) = \|u - v\|$ for all $u, v \in S$, and (b) S is closed under coordinatewise medians: for $u, v, w \in S$, the point $m(u, v, w)$ with coordinates $\text{med}(u_i, v_i, w_i)$ lies in S . Then $\Gamma(S)$ is a median graph, the median of u, v, w is $m(u, v, w)$, and*

$$d(u, v, w) = 2 \sum_{i=1}^k (\max\{u_i, v_i, w_i\} - \min\{u_i, v_i, w_i\}) \quad \text{for all } u, v, w \in S. \quad (1)$$

Proof. By (a), $\Gamma(S)$ is connected. For integers a, b, z we have $|a - z| + |z - b| \geq |a - b|$, with equality if and only if z lies between a and b . Summing over the coordinates and using (a), a point $z \in S$ lies in $[u, v]$ if and only if z_i lies between u_i and v_i for every i . An integer lies between each two of a, b, c if and only if it equals $\text{med}(a, b, c)$: if $a \leq b \leq c$, it must lie between a and b and between b and c , so it equals b , and b does lie between each two. Hence $[u, v] \cap [u, w] \cap [v, w] = \{m(u, v, w)\} \cap S = \{m(u, v, w)\}$ by (b). Finally, if $a \leq b \leq c$ then $|a - b| + |a - c| + |b - c| = 2(c - a)$; applying this in each coordinate and using (a) gives (1). $\square$

Condition (a) holds if any two points $u, v \in S$ are joined by a *monotone lattice path* in S , that is, a path in $\Gamma(S)$ along which every coordinate changes monotonically. Such a path has length $\|u - v\|$, and no path is shorter, since along each edge the value of $\|x - v\|$ changes by one.

Proof of Theorem A, except the last sentence. Place the vertices of G_1 at the points

$$3 = (0, 0), \quad 2 = (1, 0), \quad 5 = (2, 0), \quad 1 = (0, 1), \quad 0 = (1, 1), \quad 4 = (2, 1), \quad 6 = (1, 2), \quad 7 = (2, 2)$$

(Figure 1). These are the points of $S_1 = \{0, 1, 2\}^2 \setminus \{(0, 2)\}$, and the pairs of them at distance 1 are exactly the ten edges of G_1 ; so $G_1 = \Gamma(S_1)$. Two points of S_1 are joined by a monotone lattice path in $\{0, 1, 2\}^2$. If it passes through the corner $(0, 2)$, it does so between the grid neighbours $(0, 1)$ and $(1, 2)$ of the corner, and replacing $(0, 2)$ by $(1, 1)$ gives a monotone lattice path in S_1 . So (a) holds. For (b), suppose that $m(u, v, w) = (0, 2)$ for some $u, v, w \in S_1$. Then two of u, v, w have first coordinate 0 and two have second coordinate 2; one of them has both, and so equals $(0, 2) \notin S_1$, a contradiction. By Lemma 1, G_1 is a median graph.

The distance $\|u - v\|$ between points of S_1 is at most 4, with equality only for $\{u, v\} = \{(0, 0), (2, 2)\}$, as $(0, 2) \notin S_1$. Hence $\text{diam}(G_1) = 4$, the only diametral pair is $\{3, 7\}$, and the peripheral vertices are 3 and 7. By (1), $d(u, v, w) \leq 2(2 + 2) = 8$, and the vertices $1 = (0, 1)$,

$5 = (2, 0)$ and $6 = (1, 2)$ attain this bound, since in each coordinate their values range from 0 to 2. So $\text{tr}(G_1) = 8$, and $\{1, 5, 6\}$ is a triametal triple that contains neither 3 nor 7. Its distances are $d(1, 5) = 3$, $d(1, 6) = 2$ and $d(5, 6) = 3$, so it contains no diametral pair either. (On the other hand, G_1 has (Q4), as $\text{tr}(G_1) = 2 \text{diam}(G_1)$.) $\square$

Proof of Theorem B, except the last sentence. Place the vertices of G_2 at the points

$$7 = (-2, 0), \quad 6 = (-1, 0), \quad 0 = (0, 0), \quad 2 = (1, 0), \quad 1 = (2, 0), \quad 3 = (0, 1), \quad 5 = (1, 1), \quad 4 = (0, 2).$$

The pairs of these points at distance 1 are exactly the eight edges of G_2 , so $G_2 = \Gamma(S_2)$, where S_2 consists of the row $R = \{(t, 0) : -2 \leq t \leq 2\}$ and the points $(0, 1)$, $(1, 1)$ and $(0, 2)$. Two points of R are joined along R ; a point $(t, 0)$ is joined to $(0, 1)$ and to $(0, 2)$ by going along R to $(0, 0)$ and then up, and to $(1, 1)$ by going along R to $(1, 0)$ and then up; and $(1, 1)$, $(0, 1)$, $(0, 2)$ is a monotone path. So (a) holds. For (b), let $u, v, w \in S_2$ and $m = m(u, v, w)$. If at most one of u, v, w lies outside R , then $m_2 = 0$ and $m \in R$. Otherwise two of them lie in $\{(0, 1), (1, 1), (0, 2)\}$, so that $m_1 \in \{0, 1\}$ and $m_2 \in \{1, 2\}$. If $m_2 = 1$, then $m \in S_2$; if $m_2 = 2$, then two of u, v, w equal $(0, 2)$, and $m = (0, 2)$. By Lemma 1, G_2 is a median graph.

The first coordinates of the points of S_2 range over $[-2, 2]$ and the second over $[0, 2]$. So by (1), $d(u, v, w) \leq 2(4 + 2) = 12$, with equality if and only if the triple contains the only point with first coordinate -2 (vertex 7), the only point with first coordinate 2 (vertex 1) and the only point with second coordinate 2 (vertex 4). Hence $\text{tr}(G_2) = 12$, and every triametal triple consists of 1, 4, 7, which are pairwise at distance 4. One checks from the coordinates that no two vertices are at distance more than 4; so $\text{diam}(G_2) = 4$, and $d(5, 7) = \|(1, 1) - (-2, 0)\| = 4$ shows that $\{5, 7\}$ is a diametral pair and 5 is peripheral. As $5 \notin \{1, 4, 7\}$, no triple containing 5 is triametal. So $d(5, 7, z) < 12$ for every vertex z , and G_2 has neither (Q4) nor (Q4'). It has (Q3), since its only triametal triple consists of diametral pairs.

Finally, G_2 is distance-hereditary. It is obtained from the edge 02 by attaching a pendant vertex 5 at 2, then adding a vertex 3 adjacent to the two neighbours 0 and 5 of 2 (a false twin of 2), and then attaching pendant vertices 1 at 2, 4 at 3, 6 at 0 and 7 at 6. Attaching a pendant vertex and adding a twin preserve distance-heredity; this is the easy direction of [1, Theorem 1]. $\square$

So (Q4) fails even for graphs that are both median and distance-hereditary; by Theorem C, this can happen only in graphs with property (Q3), like G_2 . The example of [13] is not distance-hereditary: in its notation, the vertices b, v_2, s, w_1, x induce a path of length 4, while $d(b, x) = 2$.

4 The smallest counterexamples

The last sentences of Theorems A and B were proved by two independent exhaustive enumerations of the median graphs with at most eight vertices. The programs and their logs are in version 1.4.0 of the repository [3], directory `certificates/hko-triameter/`.

The first enumeration, a C program, runs for each $n \leq 8$ over all $2^{n(n-1)/2}$ graphs on $\{0, \dots, n-1\}$, keeps the connected ones, tests the median property from the definition for all triples, and for each median graph tests (Q3), (Q3') and (Q4), the last under both readings. Isomorphism classes come from canonical forms, over all $n!$ relabellings for $n \leq 7$ and over relabellings refined by vertex invariants for $n \leq 8$; the two agree for $n \leq 7$. The numbers of connected labelled graphs, 1, 1, 4, 38, 728, 26704, 1866256, 251548592 for $n = 1, \dots, 8$, are those of sequence A001187 of the OEIS [12], and the numbers of median graphs up to isomorphism, 1, 1, 1, 3, 4, 11, 23, 69, are those of sequence A292623 (median graphs on n nodes). No median graph with at most seven vertices lacks (Q3') or (Q4). Of the 1,008,904 labelled median graphs with eight vertices, exactly 20,160 lack (Q3') and exactly 20,160 lack (Q4). A check over all

8! bijections shows that these are the labelled copies of G_1 and of G_2 , respectively; each of G_1 and G_2 has exactly two automorphisms, and $8!/2 = 20,160$.

The second enumeration, a Python program, takes all graphs with at most seven vertices from the graph atlas of the NetworkX library and tests the median property from the definition, with Floyd–Warshall distances. For $n = 8$ it uses two facts: a median graph has no triangle, since for a triangle u, v, w the intervals $\{u, v\}$, $\{u, w\}$, $\{v, w\}$ have no common vertex; and every connected graph has a vertex whose removal leaves it connected, such as a leaf of a spanning tree. So every median graph with eight vertices arises from one of the 59 connected triangle-free graphs with seven vertices by adding a vertex adjacent to a nonempty independent set. The program forms all 1,857 such graphs and reduces the median ones up to isomorphism with NetworkX. It finds the same numbers of median graphs, no violation for $n \leq 7$, and for $n = 8$ exactly one median graph lacking (Q3') and one lacking (Q4), isomorphic to G_1 and G_2 . Finally, as (Q4) implies (Q4'), no median graph with at most seven vertices lacks (Q4'), and G_2 is the only one with eight vertices that does.

5 Distance-hereditary graphs: proof of Theorem C

Bandelt and Mulder [1, Theorem 2] proved that, for a connected graph G , the following conditions, among others, are equivalent (we keep their numbering). (i) G is distance-hereditary. (vii) For any four vertices u, v, w, x , at least two of the three distance sums $d(u, v) + d(w, x)$, $d(u, w) + d(v, x)$ and $d(u, x) + d(v, w)$ are equal. (viii) G satisfies (vii), and if the two smaller of these sums are equal, then the largest one exceeds them by at most 2. The characterisation quoted in [6, Section 3.2] is (i) $\Leftrightarrow$ (vii); we need (viii). For integers A, B, C write $\text{FP}(A, B, C)$ if two of them are equal and the third is at most their common value plus 2. This is symmetric in A, B, C , and three sums satisfy it exactly when they satisfy the requirements of (viii): if the two larger sums are equal, the third is at most their value, and if the two smaller ones are equal, (viii) bounds the largest. A connected graph G satisfies the *four-point condition* if $\text{FP}(d(u, v) + d(w, x), d(u, w) + d(v, x), d(u, x) + d(v, w))$ holds for all $u, v, w, x \in V(G)$. By the implication (i) $\Rightarrow$ (viii) of [1, Theorem 2], every connected distance-hereditary graph satisfies the four-point condition. This is the only fact about distance-hereditary graphs that we use.

The core of the proof is a statement about integers, in which a, b, c are three labels.

Lemma 2. *Let $D, e_{ab}, e_{ac}, e_{bc}, p_a, p_b, p_c$ and q_a, q_b, q_c be integers, put $e_{vu} = e_{uv}$ and $T = e_{ab} + e_{ac} + e_{bc}$, and suppose that for all distinct $u, v \in \{a, b, c\}$*

$$(E) \quad e_{uv} \leq D - 1, \quad (F) \quad \text{FP}(D + e_{uv}, p_u + q_v, p_v + q_u), \quad (N) \quad D + p_u + q_u \leq T - 1.$$

These conditions are inconsistent.

Proof. Put $s_u = p_u + q_u$ and $t_u = p_u - q_u$. Fix distinct $u, v \in \{a, b, c\}$, and let w be the third label. By (F), at least one of the following holds:

$$\begin{aligned} (A_{uv}) \quad & D + e_{uv} = p_u + q_v; & (A_{vu}) \quad & D + e_{uv} = p_v + q_u; \\ (C_{uv}) \quad & p_u + q_v = p_v + q_u \text{ and } D + e_{uv} \leq p_u + q_v + 2. \end{aligned}$$

Claim 1. *If (A_{uv}) holds, then $q_v - q_u \geq 2D + 1 - e_{uv} - e_{vw}$ and $t_u - t_v \geq 6$. Put $k = 2D + 1 - e_{uv} - e_{vw}$; by (E), $k \geq 3$. Substituting $p_u = D + e_{uv} - q_v$ into (N) for u gives $2D + e_{uv} - q_v + q_u \leq e_{uv} + e_{uw} + e_{vw} - 1$, that is, $q_v - q_u \geq k$. Substituting $q_v = D + e_{uv} - p_u$ into (N) for v gives, in the same way, $p_u - p_v \geq k$. Adding, $t_u - t_v = (p_u - p_v) + (q_v - q_u) \geq 2k \geq 6$.*

Claim 2. *If (C_{uv}) holds, then $t_u = t_v$ and $2D + 2e_{uv} \leq s_u + s_v + 4$. The equation $p_u + q_v = p_v + q_u$ says that $t_u = t_v$. Its two sides add up to $s_u + s_v$, so both equal $(s_u + s_v)/2$, and the inequality in (C_{uv}) becomes $2D + 2e_{uv} \leq s_u + s_v + 4$.*

By the two claims, for each pair the sign of $t_u - t_v$ decides which case holds: (C_{uv}) if $t_u = t_v$, (A_{uv}) if $t_u > t_v$, and (A_{vu}) if $t_u < t_v$. (For instance, if $t_u > t_v$, then (C_{uv}) fails by Claim 2, and (A_{vu}) fails by Claim 1 with u and v exchanged.) We distinguish cases by the number of pairs $\{u, v\}$ with $t_u = t_v$.

Three pairs. Adding the inequalities of Claim 2 for the three pairs gives $6D + 2T \leq 2(s_a + s_b + s_c) + 12$. By (N), $s_u \leq T - D - 1$ for each u , so $6D + 2T \leq 6T - 6D + 6$, that is, $4T \geq 12D - 6$. But (E) gives $T \leq 3D - 3$, so $4T \leq 12D - 12$, a contradiction.

Two pairs. This is impossible: if $t_u = t_v$ and $t_v = t_w$, then $t_u = t_w$ as well.

One pair, say $\{u, w\}$, and let v be the third label; so $t_u = t_w \neq t_v$. Exchanging the roles of p and q preserves (E), (F) and (N), as FP is symmetric, and replaces every t by $-t$. So we may assume that $t_u = t_w > t_v$. Then (A_{uv}) and (A_{vw}) hold, that is, $p_u = D + e_{uv} - q_v$ and $p_w = D + e_{vw} - q_v$, and hence $s_u + s_w = 2D + e_{uv} + e_{vw} - (q_v - q_u) - (q_v - q_w)$. By Claim 1, $q_v - q_u \geq 2D + 1 - e_{uw} - e_{vw}$ and $q_v - q_w \geq 2D + 1 - e_{uw} - e_{uv}$. Therefore $s_u + s_w \leq 2T - 2D - 2$, and Claim 2 for the pair $\{u, w\}$ gives $2D + 2e_{uw} \leq s_u + s_w + 4 \leq 2T - 2D + 2$, that is, $e_{uv} + e_{vw} \geq 2D - 1$. This contradicts (E).

No pair. Then t_a, t_b, t_c are distinct; call the labels 1, 2, 3 so that $t_1 > t_2 > t_3$. Then (A_{12}) , (A_{23}) and (A_{13}) hold: $D + e_{12} = p_1 + q_2$, $D + e_{23} = p_2 + q_3$ and $D + e_{13} = p_1 + q_3$. Hence $s_2 = p_2 + q_2 = (D + e_{23} - q_3) + (D + e_{12} - p_1) = D + e_{12} + e_{23} - e_{13}$, and (N) for the label 2 gives $2D + e_{12} + e_{23} - e_{13} \leq e_{12} + e_{13} + e_{23} - 1$, that is, $2e_{13} \geq 2D + 1$. This contradicts (E). $\square$

Proposition 3. *Let G be a finite connected graph that satisfies the four-point condition, let $\{x, y\}$ be a diametral pair of G , and let a, b, c be vertices such that $d(a, b)$, $d(a, c)$ and $d(b, c)$ are less than $\text{diam}(G)$. Then $\max\{d(a, x, y), d(b, x, y), d(c, x, y)\} \geq d(a, b, c)$.*

Proof. Put $D = \text{diam}(G) = d(x, y)$ and, for $u, v \in \{a, b, c\}$, $e_{uv} = d(u, v)$, $p_u = d(x, u)$ and $q_u = d(y, u)$; then $T = d(a, b, c)$. Condition (E) of Lemma 2 is the hypothesis. The three distance sums of the vertices x, y, u, v are $d(x, y) + d(u, v) = D + e_{uv}$, $d(x, u) + d(y, v) = p_u + q_v$ and $d(x, v) + d(y, u) = p_v + q_u$, so the four-point condition gives (F). If the conclusion failed, then $d(u, x, y) = D + p_u + q_u \leq T - 1$ for each $u \in \{a, b, c\}$, which is (N). This contradicts Lemma 2. $\square$

Proof of Theorem C. Let G be a connected distance-hereditary graph, or more generally a finite connected graph that satisfies the four-point condition, and let $\{a, b, c\}$ be a triametral triple that contains no diametral pair. Its three distances are at most $\text{diam}(G)$ and not equal to it. For a diametral pair $\{x, y\}$, Proposition 3 gives $\max_u d(u, x, y) \geq d(a, b, c) = \text{tr}(G)$, where u ranges over a, b, c , and the reverse inequality holds by the definition of $\text{tr}(G)$. This proves the displayed equation. If G lacks (Q3), some triametral triple contains no diametral pair, and then every diametral pair extends to a triametral triple, so G has (Q4). As (Q3) implies (Q3'), this proves Theorem C under the weak reading of (Q4).

For the strong reading, let $\{x, y\}$ be a diametral pair. If $\text{tr}(G) = 2 \text{diam}(G)$, every vertex $z \notin \{x, y\}$ extends $\{x, y\}$ (Section 2), and such a vertex exists, since a triametral triple without a diametral pair has three distinct vertices. If $\text{tr}(G) > 2 \text{diam}(G)$, the vertex $u \in \{a, b, c\}$ with $d(u, x, y) = \text{tr}(G)$ is neither x nor y , as $d(x, x, y) = d(y, x, y) = 2 \text{diam}(G)$. $\square$

Remark 4. (1) Neither property suffices alone: the graph G of [6, Figure 3] is distance-hereditary and lacks (Q3'), and the graph H of [6, Figure 3] and G_2 are distance-hereditary and lack (Q4). (2) The hypothesis on $\{a, b, c\}$ cannot be dropped: in H , of diameter 2 and triameter 6, the only triametral triple $\{a, b, c\}$ has pairwise distances 2, and the diametral pair $\{x, y\}$ does not extend. (3) Some hypothesis on G is needed: the graph of [6, Figure 2], of diameter 5 and triameter 12, has only one triametral triple $\{a, b, c\}$, with pairwise distances 4, and only one diametral pair $\{x, y\}$, which does not extend [6, Section 3.2]; as x, y are its only peripheral vertices, it has none of (Q3), (Q3') and (Q4).

Remark 5. As (Q4) holds trivially when $\text{tr}(G) = 2 \text{diam}(G)$, one might try to show that distance-hereditary graphs with $\text{tr}(G) > 2 \text{diam}(G)$ have (Q3'). This is false. Let H_{11} be the graph on $\{0, \dots, 10\}$ with the edges $01, 02, 03, 05, 14, 16, 24, 27, 34, 38, 49$ and $\{5, 10\}$: the graph $K_{2,3}$ with parts $\{0, 4\}$ and $\{1, 2, 3\}$, pendant vertices $6, 7, 8, 9$ at $1, 2, 3, 4$, and a path $0, 5, 10$. It is distance-hereditary, as $K_{2,3}$ is, and attaching pendant vertices preserves the property [1, Theorem 1]. Its only diametral pair is $\{9, 10\}$, at distance 5; the vertices $6, 7, 8$ are pairwise at distance 4; and one checks that $\text{tr}(H_{11}) = 12 > 10$. So $\{6, 7, 8\}$ is a triametal triple without a peripheral vertex. As Theorem C predicts, (Q4) holds: $\{9, 10\}$ extends by each of $6, 7, 8$, for example $d(9, 10, 6) = 5 + 3 + 4 = 12$. Up to isomorphism, H_{11} is the only connected distance-hereditary graph with at most 11 vertices that has $\text{tr}(G) > 2 \text{diam}(G)$ and lacks (Q3') (Section 6).

6 Computations on distance-hereditary graphs

These computations are not used in the proofs; programs and logs are in the directory named in Section 4. We generated all connected distance-hereditary graphs with $2 \leq n \leq 11$ vertices, up to isomorphism, from one vertex by the one-vertex extensions of [1, Theorem 1] (attaching a pendant vertex, or adding a twin of a vertex, adjacent to it or not), discarding disconnected graphs and removing isomorphic copies with NetworkX. By [1, Theorem 1] the list is complete, and its sizes for $n = 2, \dots, 11$, namely 1, 2, 6, 18, 73, 308, 1484, 7492, 40010, 220676 (270,070 graphs in all), agree with sequence A277862 of the OEIS [12]. With distances from our own breadth-first search, we checked the four-point condition on all sets of four vertices of the 49,394 graphs with $n \leq 10$; the equation of Theorem C in all 48,108 cases consisting of a triametal triple of distinct vertices without a diametral pair and a diametral pair; and (Q3) or (Q4), and (Q3') or (Q4), for every graph with $n \geq 3$. There was no failure. In all, 5,888 of these graphs lack (Q3), 1,692 lack (Q3') and 29,263 lack (Q4), but none lacks both (Q3) and (Q4). Exactly one graph in the list has $\text{tr}(G) > 2 \text{diam}(G)$ and lacks (Q3'), namely H_{11} of Remark 5.

We also tested random connected distance-hereditary graphs, built by random one-vertex extensions of the three kinds (not uniformly distributed). In 15,000 such graphs with at most 34 vertices (6,000 with $5 \leq n \leq 22$ and 9,000 with up to 34 vertices), every graph had (Q3') or (Q4); the four-point condition, checked in the first group for $n \leq 14$, always held. In 3,300 further graphs with at most 30 vertices, the equation of Theorem C held in all 578 applicable cases, the inequality of Proposition 3 held in all 11,609,908 applicable cases, and every graph had (Q3) or (Q4). A test from the definition confirmed that G_2 , H_{11} and the graphs of [6, Figure 3] are distance-hereditary and that G_1 and the graph of [6, Figure 2] are not.

7 Formal verification

The Lean files `HKOTriameter.lean` and `HKOTriameterDH.lean` (in `lean/Research/` of version 1.4.0 of the repository [3]) prove that G_1 and G_2 are median graphs with the diameters and triameters of Theorems A and B, that G_1 lacks (Q3') and G_2 lacks (Q4) and (Q4'), as shown there, and Theorem C for finite connected graphs that satisfy the four-point condition. They use Lean 4 (version 4.33.1) and Mathlib [10] (commit `0df444a3`), and all metric notions are Mathlib's `dist`, `diam`, `eccent` and `ediam` of a `SimpleGraph`. In the namespace `HKOTriameter`, `triameter G` is the maximum of `triDist G u v w = G.dist u v + G.dist u w + G.dist v w` over all ordered triples, so that vertices may repeat; `IsTriametal G u v w`, `IsDiametral G u v` and `IsPeripheral G u` say that `triDist G u v w = triameter G`, `G.dist u v = G.diam` and `G.eccent u = G.ediam`. On finite connected graphs the last is equivalent to belonging to a diametral pair, the wording of [6] (`isPeripheral_iff_isPeripheralPair`), and Problem 1 is refuted with either definition (`not_problem1ClaimPair`). `IsMedian G` says that G is connected and any three vertices have exactly one common vertex in their three intervals (`InInterval`). In ASCII notation (the files use Unicode symbols), the other definitions and the main theorems read as follows.

```

def Question3' (G : SimpleGraph V) : Prop := forall a b c : V, IsTriametal G a b c ->
  IsPeripheral G a \ / IsPeripheral G b \ / IsPeripheral G c
def Question4 (G : SimpleGraph V) : Prop :=
  forall x y : V, IsDiametral G x y -> exists z, IsTriametal G x y z
def Question4' (G : SimpleGraph V) : Prop :=
  forall x : V, IsPeripheral G x -> exists y z, IsTriametal G x y z
def Problem1Claim : Prop :=
  forall (V : Type) [Fintype V] (G : SimpleGraph V), IsMedian G -> Question3' G
-- Problem2Claim, Problem2ClaimWeak: the same with Question4, Question4' for Question3'
theorem not_problem1Claim : Not Problem1Claim
theorem not_problem2Claim : Not Problem2Claim
theorem not_problem2ClaimWeak : Not Problem2ClaimWeak
def FP (A B C : Nat) : Prop :=
  (A = B \ / C <= A + 2) \ / (A = C \ / B <= A + 2) \ / (B = C \ / A <= B + 2)
def FourPointBM (G : SimpleGraph V) : Prop := forall u v w x : V,
  FP (G.dist u v + G.dist w x) (G.dist u w + G.dist v x) (G.dist u x + G.dist v w)
theorem question3_or_question4 [Fintype V] (G : SimpleGraph V) :
  G.Connected -> FourPointBM G -> Question3 G \ / Question4 G
def Problem3ClaimFP : Prop := forall (V : Type) [Fintype V] (G : SimpleGraph V),
  G.Connected -> FourPointBM G -> Question3' G \ / Question4 G
theorem problem3ClaimFP : Problem3ClaimFP

```

Here `Question3` is (Q3), defined like `Question3'` with `IsDiametral G a b`, `IsDiametral G a c` and `IsDiametral G b c`. `Question4` does not restrict z ; this is the weak reading, so its negation is the strongest form of the refutation. The equation of Theorem C is `extends_of_no_diametral`, and Lemma 2, for natural numbers, is `core`. Other declarations record the facts about G_1 and G_2 used above (such as `G2_fourPoint`) and about the graphs G and H of [6, Figure 3], called `F1G` and `F1H` after the label of that figure in the source of [6]: both satisfy the four-point condition, G lacks (Q3') and is not median, and H lacks (Q4). The comments in the files call Theorem C “Theorem B”, and they refer to Figures 3 and 4 of [6] as “Figure 1” and “Figure 2”, after the labels in the source of [6].

Distances in the concrete graphs are given by tables, identified with Mathlib’s distance by a certificate lemma (`dist_eq_of_certificate`). The finite checks, including the median property over all triples, use `decide` and `decide +kernel` (kernel evaluation), and `core` is a split into 27 cases closed by `omega`; there is no `sorry` and no `native_decide`. Not formalised are the theorem of Bandelt and Mulder (so the formal statement of Problem 3 assumes the four-point condition, and for G_2 only the four-point condition is formal), the uniqueness statements of Theorems A and B (the only peripheral vertices of G_1 , the only triametal triple of G_2 , and the last sentences), Lemma 1, Proposition 3 as stated, the strong reading in Theorem C, Remarks 4 and 5 beyond the facts listed above, and the computations of Section 6.

The modules build with `lake build` (exit code 0, no warnings), and an audit file, `HKOTriameterAudit.lean`, prints the definitions, the statements and the axioms of 21 main declarations: all depend only on the standard axioms `propext`, `Classical.choice` and `Quot.sound`, and `core` only on `propext` and `Quot.sound`. Both compiled modules pass `leanchecker` (exit code 0), and a replay in a clean directory, from a fresh copy of the sources with the Mathlib build taken from the local cache, repeated the build, the audit and both `leanchecker` runs with exit code 0 and the same axiom lines. These checks ran on 26 September 2026 between 07:36 and 07:38 UTC; their log is among the logs in version 1.4.0 of the repository.

8 An open question

As G_1 has (Q4) and G_2 has (Q3'), our examples leave open the analogue of Problem 3 for median graphs: does every median graph have (Q3') or (Q4)? The same question can be asked for modular graphs, in which any three vertices have at least one median; each of (Q3') and

(Q4) fails for some modular graph [6, Section 4].

Tool and computational resource disclosure

This work was carried out on 26 September 2026 (UTC) in a workflow directed by the author, using AI coding agents: Anthropic Claude Code, with the model Claude Opus 5.5. The author set the research programme, namely resolving open problems with AI agents and formal verification, and directed the agents. Claude Code agents selected the question, found the results and their proofs, wrote the verification programs and the Lean formalisation, carried out the literature searches described in the introduction, and drafted this text. An independent Claude Code agent reviewed the results before the paper was written, and a second one refereed the paper: it rechecked every proof, compared the Lean statements with the theorems, checked the MathOverflow credit and the cited theorem of Bandelt and Mulder, and recomputed all counts with its own programs. The programs, logs and Lean code are in version 1.4.0 of the repository [3]. The author is not a professional mathematician. The results rest on the written proofs, on the Lean formalisation described in Section 7 and, for the last sentences of Theorems A and B, on the computations of Section 4; Section 6 gives additional checks. AI systems are not authors of this paper; the author takes full responsibility for its content.

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